SmartCold Guard: ESP32-Based IoT Cold Chain Monitoring and Security System

1. Team Information

Team Name: 10

Team Members:

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Team Leader:
Tariqul Islam

Initial Project Area:
IoT-Based Cold Chain Monitoring and Security System

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2. Problem Statement

The transportation and storage of temperature-sensitive goods such as food products, vaccines, medicines, and biological materials require continuous monitoring and protection. However, many existing cold-chain transportation systems rely on manual inspection or expensive industrial monitoring solutions, making them difficult to implement for small and medium-scale businesses.

During transportation, several problems can occur:

- Temperature and humidity conditions may exceed safe limits, causing product spoilage.
- Unauthorized access or theft attempts may happen without immediate detection.
- The location of transported goods may not be visible in real time.
- Manual monitoring cannot provide continuous data collection or instant alerts.
- Existing commercial cold-chain solutions are often costly and require specialized infrastructure.

These challenges can lead to financial losses, reduced product quality, safety risks, and inefficient logistics management.

Therefore, there is a need for a low-cost, portable, and intelligent IoT-based monitoring system that can continuously track environmental conditions, provide location visibility, control access, and detect security threats during transportation.

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3. Users / Stakeholders

The proposed system will benefit multiple stakeholders involved in cold-chain transportation:

1. Logistics and Transportation Companies

- Monitor transported goods remotely.
- Receive alerts about abnormal conditions.
- Improve shipment reliability and reduce losses.

2. Food and Pharmaceutical Suppliers

- Ensure products remain within required temperature ranges.
- Maintain product quality during delivery.
- Provide proof of proper storage conditions.

3. Warehouse Operators

- Monitor storage conditions continuously.
- Identify environmental problems before product damage occurs.

4. Delivery Personnel

- Use secure RFID-based access control.
- Reduce unauthorized handling of sensitive goods.

5. Business Owners and Managers

- Access real-time information through a centralized dashboard.
- Make better decisions based on collected data.

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4. Current Situation

Currently, many small-scale cold transportation systems depend on:

- Manual temperature checking using handheld devices.
- Periodic physical inspections.
- Basic GPS tracking without environmental monitoring.
- Traditional locks without digital access control.

These approaches have several limitations:

Lack of Real-Time Monitoring

Temperature and humidity changes may remain unnoticed until products are already damaged.

Limited Security

Traditional storage boxes cannot identify unauthorized opening or theft attempts.

No Integrated Data Platform

Environmental data, location information, and security events are usually stored separately or monitored manually.

High Cost of Existing Solutions

Industrial IoT cold-chain systems are expensive and unsuitable for small businesses.

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5. Proposed Improvement

This project proposes an ESP32-Based Smart Cold Storage & Theft Protection System that combines environmental monitoring, GPS tracking, access control, and security detection into one portable IoT solution.

The system will use an ESP32 microcontroller as the central processing unit connected with multiple sensors:

- DHT11/ DHT22 Sensor: Measures temperature and humidity.
- NEO-6M GPS Module: Provides real-time location tracking.
- MFRC522 RFID Module: Allows authorized access only.
- LDR Sensor: Detects unauthorized opening of the storage box.

- HC-SR04 Ultrasonic Sensor: Detects object presence and box status.
- Buzzer: Provides local security alarms.

Instead of using third-party platforms such as Blynk, the project will develop a custom web-based IoT dashboard. The dashboard will display:

- Live temperature and humidity readings.
- Real-time GPS location.
- Access history.
- Security alerts.
- Device status.
- Historical monitoring data.

The ESP32 will transmit collected sensor data to the custom dashboard through an IoT communication protocol, allowing users to monitor and manage the system from any location.

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6. Expected Value

The proposed system will provide the following benefits:

Reduced Product Loss

Continuous monitoring helps prevent spoilage caused by unsuitable environmental conditions.

Improved Security

RFID authentication and theft detection mechanisms reduce unauthorized access.

Real-Time Visibility

Users can track shipment location and environmental conditions remotely.

Affordable IoT Solution

The system provides features similar to industrial solutions at a lower cost.

Better Decision Making

Collected data allows users to analyze transportation conditions and improve logistics operations.

Scalable Platform

The custom dashboard architecture allows future expansion, including:

- Multiple storage boxes.
- Cloud data analytics.
- SMS/GSM notifications.
- AI-based condition prediction.

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7. Project Objectives

The main objectives of this project are:

1. To design a portable IoT-based cold-chain monitoring device.
2. To monitor temperature and humidity continuously.
3. To provide real-time location tracking during transportation.
4. To implement secure RFID-based access control.
5. To detect unauthorized opening and theft attempts.
6. To develop a custom monitoring dashboard for real-time visualization.
7. To create an affordable alternative to expensive commercial cold-chain systems.

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8. Project Scope

The project focuses on developing a prototype system capable of:

- Collecting environmental sensor data.
- Processing information using ESP32.
- Sending data to a custom dashboard.
- Generating alerts during abnormal situations.
- Providing users with remote monitoring capabilities.

Future improvements may include:

- Shock and orientation detection using MPU6050.
- GSM-based alerts using SIM800L.

- Offline data storage using SD cards.
- AI-based prediction of temperature failures.
- Battery optimization for long-term operation.